Normal: Less than 120/80 mm Hg;

Elevated Blood Pressure : Systolic between 120-129 and diastolic less than 80 mm Hg

Stage 1 Hypertension : Systolic between 130-139 or diastolic between 80-89 mm Hg

Stage 2 Hypertension : Systolic atleast 140 or diastolic at least 90 mm Hg

Hypertensive crisis: Systolic over 180 and/or diastolic over 120 mm Hg.

When the cause for hypertension is unknown, it is called **"essential or idiopathic or primary hypertension".** Hypertension of unknown causes is present in a majority of hypertensive individuals. In a small percentage of individuals, it is caused because of some other underlying health problem which may include kidney disease and diseases of the adrenal gland. This is referred to as **"secondary hypertension".**

Heredity and obesity are predisposing factors in hypertension. Smoking and stress also contribute to hypertension



Fig 11.1 Sphygmomanometer used in the measurement of blood pressure

11.2 Risk factors for hypertension

Factors responsible for hypertension are discussed below:

- **Heredity:** Children of hypertensive parents tend to have a higher risk of developing hypertension at an early age. This strongly suggests that there is a genetic basis for developing high blood pressure.
- **Age:** The prevalence of hypertension increases with age. Increase in blood pressure with age may be due to the blood vessels becoming more rigid and less elastic as one ages.
- **Gender:** Higher prevalence of hypertension has been found among males from adolescence till 45 years of age. But after the age of 45, mean blood pressure values are higher in women.
- **Obesity:** Obesity is an independent risk factor for CVD and hypertension. The severity of hypertension increases with increasing Body Mass Index also known as BMI (BMI = Weight in Kg / Height in m^2). The risk of hypertension in individuals with abdominal obesity is higher than those with normal waist circumference.



Fig 11.2 Central obesity increases the risk of hypertension

- **Physical activity:** A person who is physically inactive is prone to hypertension. Regular physical activity lowers blood pressure and improves cardiovascular health. It also helps in maintaining a healthy body weight which is an important way to control blood pressure.
- **Stress:** Chronic emotional and mental stress leads to high blood pressure.

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